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***"Transforming enterprises: Unveiling digitalization patterns
in Egypt, Morocco and Jordan"***

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"وَمَا تُوَفِّيهِ إِلَّا بِاللَّهِ"

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Abstract

Digital transformation is the topic of the current era because of its great impact on economic development. It can enhance productivity and increase process efficiency, which in turn reflects on increasing productivity, improving profitability, and enhancing market competitiveness.

This study is set out to offer a multidimensional quantitative index to measure the level of digital transformation of Egyptian, Jordanian, and Moroccan enterprises in 2022. According to the Digital Transformation Index (DTI), enterprises can be classified into two categories "Not digitalized" and "Digitalized". In addition, we investigate the main factors affecting the level of digital transformation, and test whether or not there is a significant difference in the level of digital transformation among Egypt, Jordan and Morocco.

This is achieved by constructing the Digital Transformation Index (DTI) by applying the factor analysis to relevant variables of interest, and then rescaling values to range between 0 and 1 using the normalization technique. We apply the cluster analysis to classify enterprises into two categories (not digitalized and digitalized) and check its validity using the discriminant correspondence analysis. Then, we apply the binary logistic model, in which the response variable is the Digital Transformation Index (DTI), and decision tree learning to identify the main factors affecting the level of digital transformation.

A Digital Transformation Index (DTI) is presented. The closer the DTI value to 1, the more digitally transformed the enterprise is. We find that the main activity of the enterprises, the educational level of the owner, the enterprise's human capital, spending on research and development, the cost of the internet as a share of total cost, and the percentage of educated workers have a significant effect on the level of digital transformation. In studying the effect of countries on the level of digital transformation, we find that they have no significant effect on the level of digital transformation.

Key words: Digital Transformation. Factor analysis. Cluster analysis. Binary logistic model. Decision tree learning.

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List of Acronyms

Acronym	Full Name
AUC	Area Under the Curve
BCSS	Between clusters sum of squares
DBS	Digital Business Strategies
DTI	Digital Transformation Index
ERF	Economic Research Forum
EU	European Union
GDP	Gross Domestic Product
OECD	The Organization for Economic Cooperation and Development
PCA	Principal Component Analysis
ROC	Receiver Operating Characteristic
SMEs	Small and Medium-Sized Companies
TOPSIS	The Technique for Order Preference by Similarity to an Ideal Solution
VIF	Variance Inflation Factor
WCSS	Within- Cluster Sum of Squares

Chapter (1)

Introduction

1.1 Overview

The coronavirus pandemic has changed the course of the entire world economically, socially, politically, and culturally. During this time, the need for digital transformation has emerged; this strategy has evolved further for various governmental and private enterprises. This new change has become clearly accepted by clients and service providers. The telecommunications sector is now more than just traditional communication and information search; it is also the backbone of the use of digital data, content, and applications by individuals, governments, and companies to ensure the continuity of economic and social activity under social distancing and complete closure.

The implementation of the digital transformation strategy was hampered by several challenges prior to the coronavirus pandemic, especially in developing countries. Most notably, the culture, the acceptance of the idea of digital transformation, the possibility of providing services, and conducting transactions electronically without direct face-to-face interaction between service providers and recipients. It is difficult to adopt a digital transformation strategy by a department that does not believe in the importance of digital transformation and offers it to customers who are unaware of the importance of that path. In addition to the challenges enterprises face in preparing for digital transformation, they also face the inavailability of robust and high-quality infrastructure that enables them to conduct the same transactions with high efficiency, as well as the existence of large investments and strong financial capabilities. That development led researchers to talk about the possibility of applying a corporate digital transformation strategy, measuring it through quantitative indicators, and comparing between those companies through the level of digital transformation they have and the extent to which it affects their performance and annual revenue.

Once we talk about digital transformation as a strategy, we should address a fundamental difference between these terms: Digitalization and digitization. The term digitization refers to the conversion of information from analogue to digital by converting paper documents into electronic files and using databases and information systems to regulate data, while digital transformation refers to the use of digital technology to change the way it works. Therefore, we can say that the term digital transformation is broader and more comprehensive than the term digitization.

Digital transformation aims to improve productivity and increase process efficiency, which in turn reflects on increasing productivity, improving profitability, enhancing market competitiveness, and contributing to maintaining cybersecurity through high-security protection of data and information circulating online. Creativity and innovation are also unleashed; as digital

technology can allow the development of new and innovative products and services, it can also accelerate and facilitate creative processes.

1.2 Study problem:

It is noticed that there is no uniform measure for the level of digital transformation by which enterprises in Egypt, Jordan and Morocco can be classified. We also need to determine the main factors affecting the level of digital transformation and study the extent of the impact of different countries on the level of digital transformation.

1.3 Objectives:

The objectives of this study could be summarized, as follows:

1. Constructing a multidimensional quantitative index to measure the level of digital transformation in enterprises in 2022.
2. Clustering enterprises according to their performance in digital transformation.
3. Investigating the main factors affecting the digital transformation.
4. Testing whether or not there is a significant difference in the level of digital transformation among Egypt, Jordan and Morocco.

1.4 Literature reviews:

There are a lot of studies and papers related to digital transformation in several aspects, and here, we review some of these studies on indicators used to measure digital transformation:

1.4.1 Local and Arab studies:

- The Digital Government Authority of Saudi Arabia (2023) released an executive report to study the measurement of digital transformation for government agencies. The study relied on descriptive analysis to measure and assess progress and the shift towards effective use of digital technology in government entities. The study reached several key findings in measuring the digital transformation of government entities. These results included the advancement of entities in their creative models of digital transformation, as well as improving the performance and effectiveness of the entities.
- Mohammed et al (2022) built a quantitative index to measure the level of digital transformation in business enterprises, which aimed to help practitioners measure the levels of progress in adopting and applying digital transformation technologies with a view to the sustainability of enterprises. The various actions of business enterprises were categorized as being digitally transformed into seven axes: infrastructure, organizational culture, financial

capacity, human resources, digital transformation techniques, digital business, and integration and development. This study relied on both inductive¹ and deductive² methodologies, as well as the mixed strategy that combines quantity and description in order to interpret the results of quantitative data through a descriptive interpretation. After reviewing the axes of the actions underlying digital transformation, the study developed a set of indicators for each component of each axis and combined them together into a composite indicator.

1.4.2 Foreign studies:

- Brodny and Tutak (2021) conducted a study of the level of digital transformation in European Union (EU) member states. The study aimed to identify the level of digital transformation of European companies according to their size: small, medium, and large companies. The study aimed to identify the similarities among EU countries in the level of digital transformation of their companies taking into account the size of these companies.

The analysis used both:

1. Application preference style by resembling the Ideal Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS).
2. Kohonen's artificial neural networks.

The conclusions of this study included:

1. Identifying the main determinants of the European companies' digital transformation index.
2. Dividing EU countries based on the level of digital transformation into four levels: high, high average, low, and low average.

- Brodny and Tutak (2022) estimated the level of digital transformation and the use of innovative technology among companies in EU member states. The research also aimed to identify the relationship between the digital transformation index and the economic parameters of these countries.

Analytical methods were used in the study. Non-parametric methods, such as the Kendal and the Spearman coefficients were used to determine the relationship between the digital numbering index and economic parameters.

Key findings on the levels of digitization in EU institutions included identification of the digital techniques used, identification of the levels of their implementation, the establishment of relationships between these technologies, and their impact on economic parameters within

¹ Inductive research uses specific observations and patterns to come up with new theories.

² Deductive research starts with a theory or hypothesis and tests it through observations.

the EU27 countries. Moreover, this indicated a positive and significant relationship between internet use, financial development, and economic growth. It also highlighted the impact of financial development and increased investment in communications technologies on per capita Gross Domestic Product (GDP) growth.

- Kharlamov and Parry (2020) proposed a systematic method of analyzing the impact of servitization on the financial and economic situation, as well as aiming to quantify and synergize service and digitization by examining their relationship to financial and economic outcomes.

The study used standard economic analysis to assess the impact of service and digitization on companies' financial and economic performance, which were classified into four categories based on their descriptions: pure product companies (P companies), digital companies (D companies), serviced companies (S companies), and digitized and serviced companies (DS companies).

The study concluded that there was no significant difference between productivity and profitability between P and D companies. However, both S companies and DS companies showed greater productivity than P companies, and the profitability of DS companies was greater than that of P companies, while S companies were less profitable than PS companies.

- Mammadli and Klivak (2020) measured the level of digital transformation nationwide and its economic and social impact on its growth by building a digital transformation index, which was applied to the Organization for Economic Cooperation and Development (OECD) countries' data from 2009 to 2017. The index consisted of three main dimensions: the level of development of infrastructure, education, and skills.

The Cobb-Douglas model was used on OECD countries' data during the nine years referred to above. The state's gross domestic product (GDP) was used at the time point as a subordinate variable with the use of the two main components of the model: investment and the proportion of employees aged 15 years or older. This was in addition to the digital transformation indicator as well as the state's GDP in the year preceding the year specified. This was to measure the extent to which this affected the state's domestic product and thus the overall economic level.

The most significant finding was the moral impact of the digital transformation of the economy on the state's economic growth.

- Mies et al (2023) conducted a study with the objective of classifying digital business strategies (DBS), as well as identifying and classifying different types of DBS according to the following four types: non-digital reactor, analyzer, digital opportunist, and digital product to assess their impact on the company's performance.

Cluster analysis was used. The study found that four types of business strategies were identified and assessed for their impact on the company's performance. Each type showed different characteristics. The study also assumed that a digital product could outperform other species, while digital opportunism lacked strategic plans and acted unpredictably.

- Stich et al (2023) undertook a study to determine the current status of Small and Medium-Sized Companies (SMEs) and the necessary measures to put those companies on the digital transformation map. Four measures were identified to measure SMEs' digital transformation: resources (such as assets, machinery and infrastructure), information systems, culture and organizational structure.

This study relied on theoretical reasoning for selecting supported SME models with the aim of expanding the model under development.

The study found a correlation between the global digitization metrics of SMEs and the digital transformation through the analysis of eleven companies.

- Stoyanova (2022) measured digital transformation in Bulgaria. It can be measured by the level of corporate internet connectivity, the number of employees that were able to use the internet and computers, the availability of a company's website or not, and some other indicators.

The study relied on descriptive analysis and some of its methods, such as bilateral consensus tables to analyze the relationships between the above variables.

One of the most important findings of the study was that the most widely used technological means by small companies was the use of the internet and computers. For small businesses, there was a weak relationship between having a company's own website and doing transactions through e-commerce. Finally, the relative contribution of small companies to e-payment services and big data analysis was four times less than that of large companies. This was due to the disparity between large and small companies in the use of modern technological means, as well as to large activity volumes and large numbers of workers.

- Yakymova et al (2022) created a composite index to measure the level of digital transformation in order to allow countries to be classified by the level of digital transformation of the utility industry, as well as identify strengths and weaknesses in the process in the countries to which the analysis was applied. The study was applied to 34 European countries based on 31 indicators from the European Commission database.

When conducting multivariate analysis for basic indicators, the correlation matrix analysis was limited to be used in the principal component analysis (PCA). After one pair of highly correlated indicators was excluded, a final set of 31 indicators was reached. Linear programming was used in calculating each state's composite indicator. One of its constraints was that the composite indicator of another state under analysis should not exceed one when using optimal weight values for the state under analysis.

The study found that through the development of the Industry Digital Transformation Index, the industry's digital transformation and the techniques used to build a composite indicator could be better understood, which also benefits industry policymakers because it refers to the strengths and weaknesses of the digital transformation of facilities.

Chapter (2)

Methodology

2.1 Data and software used

Data collected through the survey of enterprises' digitization, which was conducted by the Economic Research Forum (ERF) in 2022 is used to construct the Digital Transformation Index.

The target population of the survey was all enterprises of different sizes that started to conduct their activities before 2022. The sample was designed to represent both electronic and non-electronic companies. Electronic companies were defined as those companies that used the internet to conduct transactions, such as buying and selling goods or services, a month before interviewing them. Interviews were conducted by telephone. Therefore, the framework was a list of telephone numbers of companies that started their activities before 2022. The questionnaire was designed and implemented to understand the impact of digitization on the company's productivity, and includes questions about the characteristics of companies.

The total sample size used in our study is 948 companies (306 Egyptian enterprises, 311 Jordanian enterprises, and 331 Moroccan enterprise).

All statistical analysis is performed using R 4.4.0.

2.2 Methodology

2.2.1 Indicator construction:

There are multiple techniques for constructing indices, such as factor analysis and aggregation techniques. This section represents an overview of these techniques.

I. Factor analysis:

Factor analysis is a data reduction technique used to uncover underlying structures in a dataset. It helps to identify groups of variables that are highly interrelated and can be used to explain a common underlying theme. It is also used to construct an indicator.

Factor analysis is a technique to investigate whether a number of variables of interest, $X_1, X_2, \dots, X_p$, are linearly related to a smaller set of latent (unobservable) factors, $F_1, F_2, \dots, F_m$.
 $\forall m < p$

$$\mathbf{X}_{(p \times 1)} = \mathbf{L}_{(p \times m)} \mathbf{f}_{(m \times 1)} + \mathbf{u}_{(p \times 1)}$$

Where:

$\mathbf{X}$: is the vector of original variables.

$\mathbf{L}$: is the factor loadings matrix.

$\mathbf{f}$: is the vector of the common factors.

$\mathbf{u}$: is the vector of unique factors.

It is assumed that a measurement error has a constant variance and is on average zero $E(e_i) = \text{zero}$, $V(e_i) = \sigma^2$, and there is no association between the factor and measurement error $\text{cov}(F, e_i) = \text{zero}$, and no association between errors $\text{cov}(e_i, e_j) = \text{zero}$. It implies that the covariance between any two observed variables is determined by their factor loadings on the common factors, and the variance of each observed variable is the sum of two components: communality and uniqueness.

In this report, we determine the number of factors to retain using different approaches, for example: The scree plot is a line graph of the values of the eigenvalues against the eigenvalue order. Then we extract the factors using several methods, for example: The principal factor method, in which we assume initial estimates of the unknown communalities to extract the common factors, and then we update these communalities and redo the process until the results converge. The extracted factors should be orthogonal and ordered in terms of their importance. Then we can rotate the factors (if needed) to make them more interpretable. Finally, we estimate the values of the factors for different sample units.

The polychoric correlation matrix is used to extract the common factors instead of the usual reduced or unreduced correlation matrix since the variables of interest are categorical variables. The coefficients of a polychoric correlation matrix are called tetrachoric correlation coefficients, and are based on the assumption that there are underlying continuous latent variables that generate the observed data. The polychoric correlation matrix is based on the observed frequencies or proportions in each cell of a contingency table created from the variables. It provides a measure of the strength and direction of the relationship between the variables, similar to the Pearson correlation matrix for continuous variables. It is useful in exploratory factor analysis, structural equation modelling, or any analysis that involves assessing relationships between variables. Distefano et al (2009).

The Bartlett factor scores method:

With Bartlett's approach, only the shared (i.e., common) factors have an impact on factor scores. The sum of squared components for the "error" factors (i.e., unique factors) across the set of variables is minimized, and resulting factor scores are highly correlated to their corresponding factor and not with other factors. However, the estimated factor scores between different factors may still correlate.

This method to get the scores of the factors is based on using a multiple regression technique to estimate the factor scores for each individual in the dataset, in which the observed variables are considered independent variables and the estimated factor loadings for each variable are considered the regression weights. In the end, the resulting regression coefficients represent the factor scores for each individual.

Bartlett factor scores are computed by multiplying the row vector of observed variables by the inverse of the diagonal matrix of variances of the unique factor scores and the factor pattern matrix of loadings. The resulting values are then multiplied by the inverse of the matrix product of the matrices of factor loadings and the inverse of the diagonal matrix of variances of the unique factor scores. Then the factor scores are normalized to range from 0 to 1, so that we can construct our indicator. The Bartlett factor scores method using the multiple regression technique can be written, as follows:

$$\hat{F}_{(1 \times m)} = X_{(1 \times p)} M_{(p \times m)}$$

Where:

P: is the number of observed variables. **m**: is the number of factors.

X: is the row vector of *p* observed variables.

M: is the matrix of regression weights for the *m* factors on *p* observed variable.

F: is the row vector of *m* estimated factor scores.

II. The aggregation techniques:

The dimensions are constructed using the aggregation of a group of binary variables. Each dimension is normalized using min-max normalization.

$$D_{normalized} = \frac{D - \min_D}{\max_D - \min_D}$$

These dimensions are combined into one composite indicator using either geometric or linear aggregation, respectively.

$$\prod_{i=1}^n D_i^{w_i} \qquad \sum_{i=1}^n w_i * D_i$$

Where:

D: is the dimension.

n: is the number of dimensions.

w_i: is the weight assigned to each dimension I, subject to $\sum_{i=1}^n w_i = 1$.

Blancas (2013).

In our study, we use the factor analysis technique in constructing the Digital Transformation Index (DTI).

2.2.2 Cluster analysis:

- **K -modes clustering:**

The K-modes algorithm is an extension of the K-means algorithm specifically designed for clustering categorical data. Unlike the K-means algorithm, which is suitable for numerical data, the K-modes algorithm handles categorical data by using a different dissimilarity measure and updating cluster modes based on the frequency of attribute values.

An overview of how the K-modes algorithm works:

1. Initialization: The algorithm starts by randomly selecting initial cluster centers (modes) as seeds. This step is crucial, as the clustering results can heavily depend on the choice of initial cluster centers.
2. Distance measure: Instead of using the Euclidean distance as in the K-means, the K-modes algorithm evaluates the dissimilarity between categorical data objects using a match dissimilarity measure or Hamming distance.
3. Update cluster modes: The algorithm updates the cluster modes based on the frequency of attribute values in each cluster. This frequency-based method helps in determining the most representative values for each cluster.
4. Cost function minimization: The K-modes algorithm aims to minimize the cost function, which is defined based on the dissimilarity between data objects and their respective cluster centers.
5. Cluster formation: By iteratively updating the cluster modes and reassigning data objects to the nearest cluster based on the dissimilarity measure, the algorithm forms clusters of categorical data objects.

Sharma and Gaud (2015).

The dissimilarity measure (Hamming Distance) and cost function play crucial roles in the K-modes algorithm for clustering categorical data. Where The Hamming distance is a measure of dissimilarity between two categorical data objects. It calculates the number of attributes in which the two objects differ.

For two categorical data objects x and y with m attributes, the Hamming distance is computed as:

$$d(X_i, X_l) = \sum_{j=1}^m \delta(x_{ij}, x_{lj})$$

Where

$$\delta(x_{ij}, x_{lj}) = \begin{cases} 0 & , \text{ if } x_{ij} = x_{lj} \\ 1 & , \text{ if } x_{ij} \neq x_{lj} \end{cases}$$

Where x_i and x_l represent two observations. X_i and X_l represent two vectors for all m attributes for observation i and l respectively.

The Hamming distance provides a simple and effective way to quantify dissimilarity between categorical data objects, giving equal weight to each attribute. Huang and Ng (2003).

- **Cost function:**

The cost function in the K-modes algorithm is used to evaluate the overall dissimilarity between data objects and their respective cluster centers. Minimizing this cost function leads to optimal cluster formation.

Let Z be a set of categorical data objects described by categorical attributes, $A_1, A_2, \dots, A_m$. While the above is used because the dissimilarity determines categorical data objects, the cost function becomes:

$$C(Q) = \sum_{i=1}^n d(Z_i, Q_i)$$

Where:

Z_i represents the i th element of the categorical data objects set Z .

Q_i is the nearest cluster center to Z_i .

The algorithm iterates through the data objects, assigning each object to the cluster with the nearest mode (center) based on the dissimilarity measure (Hamming distance). The cost function aggregates the dissimilarities for all data objects, reflecting the overall clustering quality. By updating the cluster modes and reassigning data objects iteratively to minimize the cost function, the K-modes algorithm optimizes the clustering process for categorical data.

- **Determining number of clusters & validating:**

The elbow method is a technique used in clustering analysis to determine the optimal number of clusters in a dataset. Here is how the elbow method works:

1. Cost function calculation: The elbow method starts by calculating a cost function denoted as J :

$$J = \sum_{i=1}^k \sum_{x \in C_i} |x - C_i|^2$$

Where:

x is the element of cluster C_i .

k is the number of clusters $|C_i|$.

The cost function measures the sum of squared distances between data points and their respective cluster centroids. This is known as the Within-Cluster Sum of Squares (WCSS). It is calculated for different numbers of clusters (k).

2. Elbow curve visualization: The next step involves plotting the cost function J against the number of clusters k . As the number of clusters increases, the cost function typically decreases because the data points are more closely clustered. This results in a curve that resembles an elbow, as shown below using iris data:

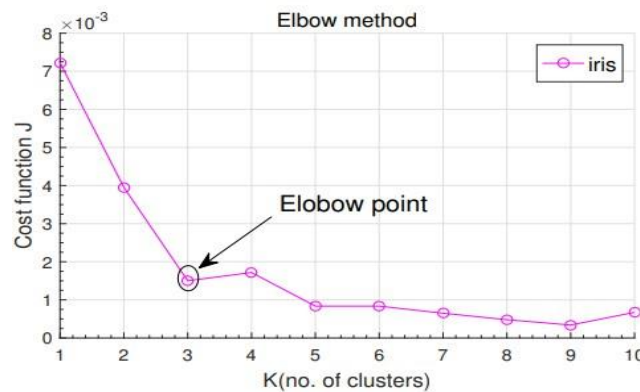


Figure (2.1): Elbow method

3. Identifying the elbow point: The key concept of the elbow method is to look for the point on the curve where the rate of decrease of the cost function J significantly slows down, forming an elbow-like bend. This point is considered the optimal number of clusters for the dataset.

4. Selecting the optimal number of clusters: The optimal number of clusters is determined by selecting the value of k at the elbow point on the curve. This value represents the point where adding more clusters does not significantly improve the clustering quality, since if we continue adding clusters beyond the elbow point, the reduction in WCSS becomes marginal while the model complexity increases.

5. Application in clustering: Once the optimal number of clusters is identified using the Elbow method, it can be used to perform clustering analysis on the dataset with the appropriate number of clusters. Liu and Deng (2021).

The silhouette score is a measure of how similar an object is to its own cluster (cohesion) compared to other clusters (separation). It ranges from -1 to 1, where a high value indicates that the object is well matched to its own cluster and poorly matched to neighboring clusters. If most objects have a high value, then the clustering configuration is appropriate. If many points have a low or negative value, then the clustering configuration may have too many or too few clusters.

The silhouette score is calculated for each sample and provides a way to assess the quality of the clustering. It takes into consideration the mean intra-cluster distance and the mean nearest-cluster distance for each data point.

The silhouette score for a sample is

$$\frac{b - a}{\max(a, b)}$$

Where:

a is the mean intra-cluster distance.

b is the mean nearest-cluster distance.

Samuel and Mironov (2018).

2.2.3 Binary logistic model:

Binary logistic model is a type of regression model used to predict the probability of an event occurring based on one or more explanatory variables; these variables could be categorical or quantitative. The response variable Y should be binary with two categories coded as 1 for success and 0 for failure. The equation of binary logistic model can be written, as follows:

$$\text{logit}(\pi) = \alpha + \sum_{i=1}^k \beta_i x_i$$

or by another way can be written as follows:

$$\pi(Y) = \frac{e^{(\alpha + \sum_{i=1}^k \beta_i x_i)}}{1 + e^{(\alpha + \sum_{i=1}^k \beta_i x_i)}} ; \quad Y = \begin{cases} 1; \text{ if it is a success.} \\ 0; \text{ if it is a failure.} \end{cases}$$

The estimates for the model parameters are obtained using the maximum likelihood method.

- **Binary logistic model assumptions:**

To be confident about the results of the binary logistic model, we should first check the satisfaction of the assumptions of the model, which can be illustrated, as follows:

- Linearity of the logit: The logistic model does not imply a linear relationship between the dependent and independent variables, as there is a S-shape relationship between each variable and the probability of the response variable but it requires that the independent variables are linearly related to the log odds of the response variable, and we can check it by plotting the predicted probabilities against the continuous predictor variables.
- Independence of observations: The observations should not come from repeated measurements or matched data.
- Absence of multicollinearity: There is no multicollinearity among the predictor variables by using the Variance Inflation Factor (VIF) or correlation matrices.
- Absence of outliers: There are no outliers or influential points.

- **Goodness of fit:**

To know whether the model fits the data well or not, we can detect that by applying multiple methods of goodness of fit for the binary logistic model:

- **Pearson's Chi-squared test or Likelihood Ratio test:** They have an approximate chi-squared distribution. When all explanatory variables are categorical, the chi-squared approximation is valid. The high p-value indicates a very good fit.
- **ROC (Receiver Operating Characteristic) curve:** The curve draws the values of sensitivity against the values of $(1 - \text{specificity})$ at all possible cut-off points. Using the AUC (Area Under the Curve), we can detect the quality of the performance of the model. If the AUC is higher than 0.5, this indicates that the model has good predictive power but if the AUC equals 0.5, this indicates that the model does not have good predictive power and does not differ from the intercept model, which classifies the observations randomly.
- **Classification table:** It describes the predictive power of the model, which contains the percentage of correctly classified as positive, which is called sensitivity and the percentage of correctly classified as negative, which is called specificity. It also contains the percentage of overall correct classification, which indicates that the model has acceptable predictive power if it is greater than 60% but if it is equal to 50%, it indicates that there is a random classification.

2.2.4 Decision tree:

Decision tree is a machine learning technique that is mainly used to classify the target variable based on the input variables. The decision tree can be divided into a classification tree if the variable we wish to predict is categorical, and a regression tree if that variable is continuous. A classification tree can be used with a set of if-then logical conditions to classify the target variable; it is also used to determine the most important variables in classification.

The main parts of a decision tree are:

- **Nodes**: test the value of a certain attribute.
- **Leaves**: terminal nodes by splitting the nodes that predict the outcome and the decision based on the majority of voting.
- **Edges**: correspond to the outcome of a test and connect to the next node or leaf.

The decision tree can be conducted in two stages:

- 1- **Recursive partitioning stage**: split the observations repeatedly into two or more branches to reach maximum homogeneity within new parts.
- 2- **Pruning the tree**: simplifies the tree by pruning peripheral branches to avoid over fitting.

If one of the classes occurs for each observation, then the data is called pure. There are multiple measures for impurity, such as entropy, which can be written, as follows:

$$Entropy = - \sum_{i=1}^k p_i * \log_2(p_i) \quad \forall i = 1, 2, \dots, n.$$

Entropy measures the variation for each variable in the tree, so we try all possible splits to reach the split with the minimum entropy. For the categorical variables, the model tries all possible splits and chooses the best in terms of entropy. For the quantitative variables, the values of the continuous variable are sorted in an increasing order, then we split based on the midpoint, and the best is chosen in terms of the entropy.

Suthaharan (2015).

Chapter (3)

Data description

In this chapter, we show a descriptive analysis of the variables included in our study. Our sample of enterprises is consisted of 32.3% Egyptian enterprises, 32.8% Jordanian enterprises, and 34.9% Moroccan enterprises, which makes us rule out the existence of a confounding variable as the distribution in each country is almost identical. The head office of most enterprises exists in urban areas (94.7%).

We also notice that the majority of enterprises included in our study are owned by the private domestic sector (98%), and only 2% of enterprises are owned by the private foreign sector. In our sample, the number of male owners is higher than the number of female owners. Most owners in our sample have a university education or above.

The majority of enterprises work in the vehicle industry, do not spend money on research and development, and do not introduce new products to the main market. According to the Digital Transformation Index, we find out that the majority of enterprises included in our study are not digitalized (65.3%).

Next, we study the association between the Digital Transformation Index (DTI) and other related variables, as follows:

- **Association between the DTI and other related variables:**

Table (3.1): Association between the DTI and other related variables.

Variable	Measure of association	p-value
Country	$\hat{U} = 0.0024$	0.1333
Human capital	$\hat{\gamma} = 0.22$	4.55e-05 ***
Main activity	$\hat{U} = 0.006$	0.04341 *
Region of the head office	$\hat{U} = 0.011$	0.00689 **
Age of the enterprise	$\hat{\gamma} = 0.113$	0.07138 *
Education of the owner	$\hat{\gamma} = 0.26$	5.393e-08 ***
Percentage of educated workers	$\hat{\gamma} = 0.27$	4.322e-07 ***
Number of competitors	$\hat{\gamma} = -0.062$	0.3402
Introducing new product	$\hat{U} = 0.017$	3.198e-06 ***
Research and development	$\hat{U} = 0.038$	8.083e-12 ***
Period of access to the internet	$\hat{\gamma} = 0.053$	0.7244
Cost of the internet	$\hat{\gamma} = 0.089$	0.1707
Using the computer	$\hat{\gamma} = 0.048$	0.1597
Digital skills for workers	$\hat{U} = 0.005$	0.008922 **
Number of power outage days	$\hat{\gamma} = 0.043$	0.4075
Number of no internet days	$\hat{\gamma} = 0.005$	0.9074

Significant codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘’ 1

Table (3.1) shows the association between DTI and other related variables. For the ordinal variables, we measure the strength and direction of association using the Gamma coefficient ($\hat{\gamma}$), and for the nominal variables, we measure the strength of association using the Uncertainty coefficient ($\hat{U}$).

With 95% confidence, there is a moderate positive association between DTI and human capital, the education of the owner, and the percentage of educated workers, which means that when the enterprise has many paid workers, the owner of the enterprise is highly educated, and the enterprise has many educated workers among its human capital, it tends to be digitalized.

With 95% confidence, there is a weak association between DTI and the main activity, the region of the head office, introducing a new product to the main market, spending on research and development, and the digital skills of workers, which means that the main activity, the region of the head office, introducing a new product, spending on research and development, and the digital skills of workers reduce the entropy of the digital transformation of the enterprise by 0.6%, 1.1%, 3.8%, and 0.5%, respectively.

Therefore, we conclude that the more related variables to the Digital Transformation Index are human capital, the education of the owner, and the percentage of educated workers.

In the next part, we show the summary statistics of the Digital Transformation Index, as follows:

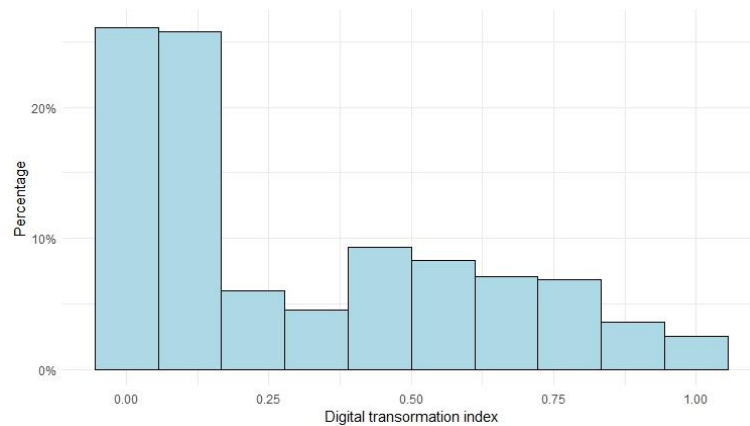


Figure (3.1): Histogram of the Digital Transformation Index.

Figure (3.1) shows that the distribution is skewed to the right. The majority of values of DTI are located on the left side of the graph.

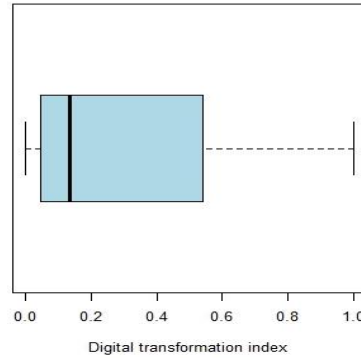


Figure (3.2): Boxplot of the Digital Transformation Index.

Figure (3.2) shows that there are no outliers. The distribution is skewed to the right.

Table (3.2): Testing the normality of the Digital Transformation Index.

Test	Statistic	P-Value
Mardia Skewness	186.643878107258	2.7887497786753e-39
Mardia Kurtosis	-4.29403540983962	1.75454472437497e-05

Table (3.2) shows that the P-value of the normality test is less than ($\alpha = 0.05$). Therefore, we reject the null hypothesis, which states that the distribution follows the univariate normal distribution.

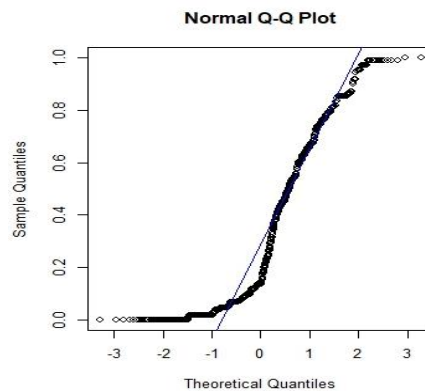


Figure (3.3): Normal Q-Q Plot.

Figure (3.3) shows that values of DTI are not normally distributed.

After applying the cluster analysis, enterprises are clustered into two categories, as follows:

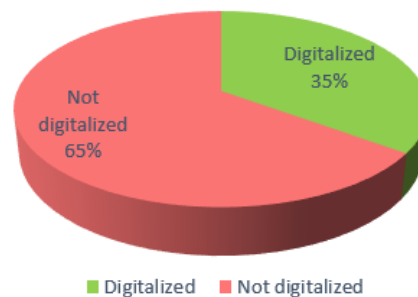


Figure (3.4): The distribution of the Digital Transformation Index.

Figure (3.4) shows that most enterprises included in the study are not digitalized.

Then, we show the distribution of the Digital Transformation Index by demographic and characteristic variables, as follows:

- **The distribution of the Digital Transformation Index by country:**

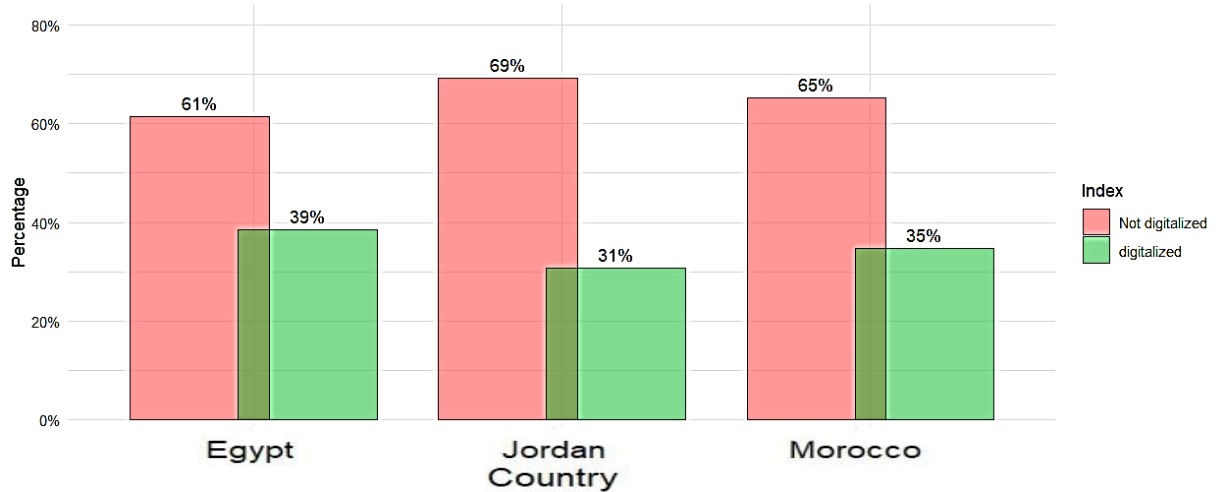


Figure (3.5): The distribution of the DTI by country.

Figure (3.5) shows that the distribution of DTI is nearly similar in the countries under study. 39% of the Egyptian enterprises are digitalized, which is the highest percentage of digital enterprises in our sample. 69% of the Jordanian enterprises are not digitalized, which is the highest percentage of non-digital enterprises in our sample. The gap between the percentage of being digitalized and not digitalized is less for Egypt than for other countries included in our study.

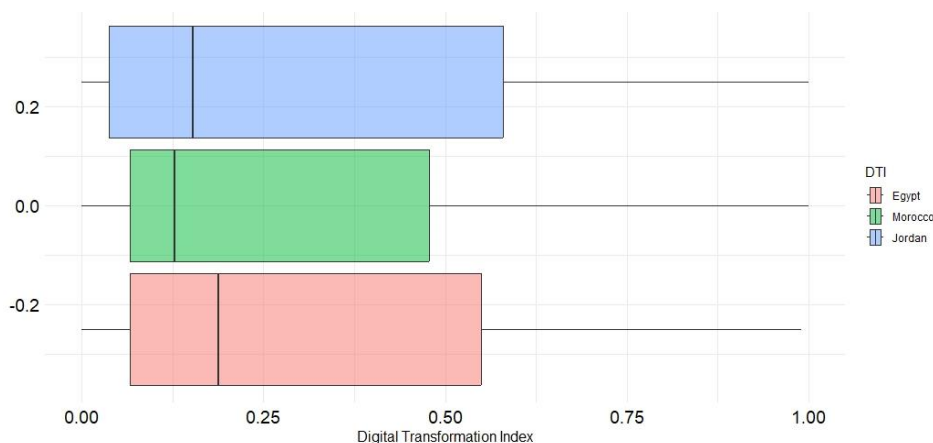


Figure (3.6): Box plots of the DTI by country.

Figure (3.6) shows that the distribution of the Digital Transformation Index by country is skewed to the right. The median value of all countries is small. Egypt has the highest median. There are no outliers.

- **The distribution of the DTI by region of the head office:**

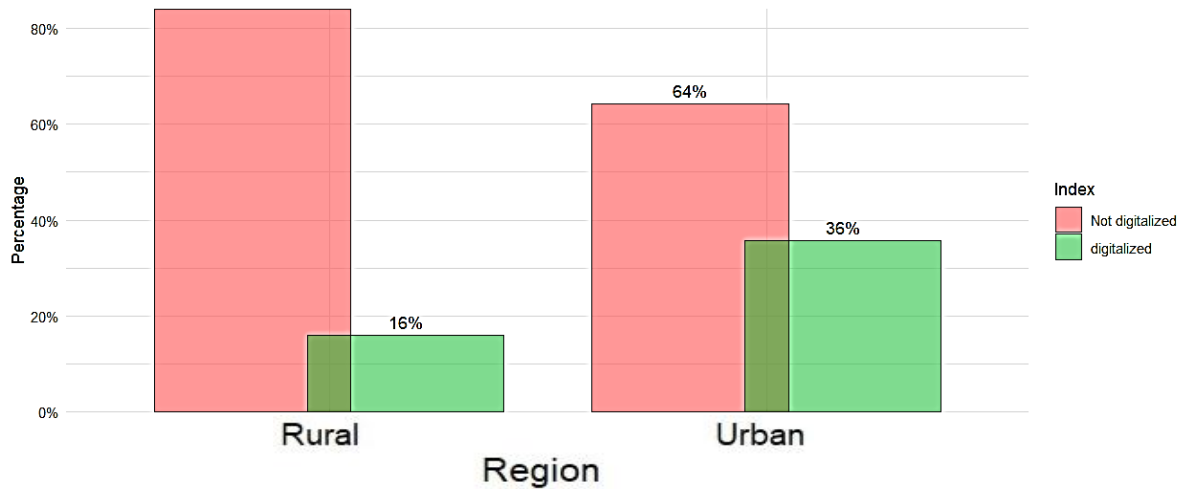


Figure (3.7): The distribution of the DTI by region of the head office.

Figure (3.7) shows that the highest percentage of being "digitalized" is for enterprises whose head office is located in urban areas. While the highest percentage of being "not digitalized" is for enterprises whose head office is in rural areas.

- **The distribution of the DTI by human capital:**

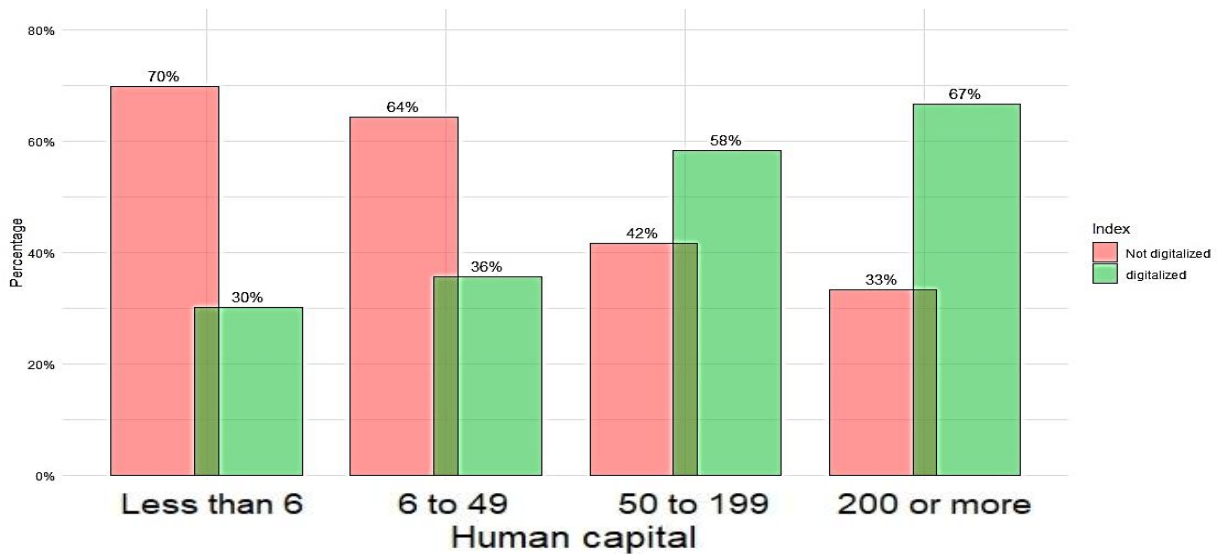


Figure (3.8): The distribution of the DTI by human capital.

Figure (3.8) shows that the percentage of being "digitalized" is higher than "not digitalized" for enterprises that have large human capital and vice versa for enterprises that have small human capital.

- **The distribution of the DTI by sex of the owner:**

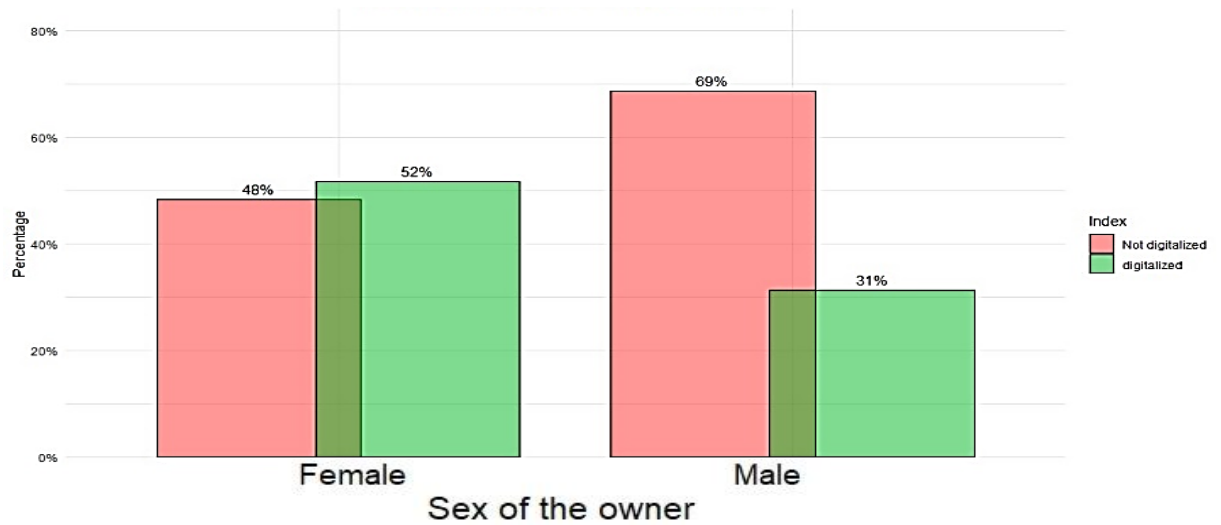


Figure (3.9): The distribution of the DTI by sex of the owner.

Figure (3.9) shows that the percentage of being "digitalized" is higher for female owners. The gap between the percentage of being "digitalized" and "not digitalized" is less for female owners than male owners in our study.

- **The distribution of the DTI by education of the owner:**

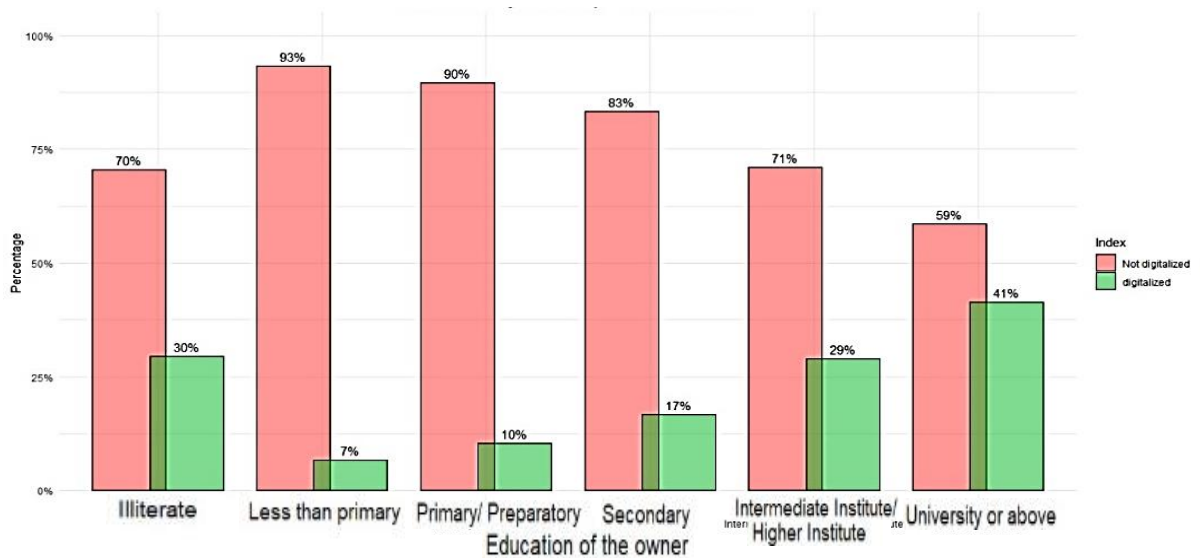


Figure (3.10): The distribution of the DTI by education of the owner.

Figure (3.10) demonstrates that 41% of enterprises with owners who have completed their education at a university or above are digitalized. The gap between the percentage of being "digitalized" and "not digitalized" for the category "university or above" is not large compared to other categories.

- **The distribution of the DTI by percentage of educated workers:**

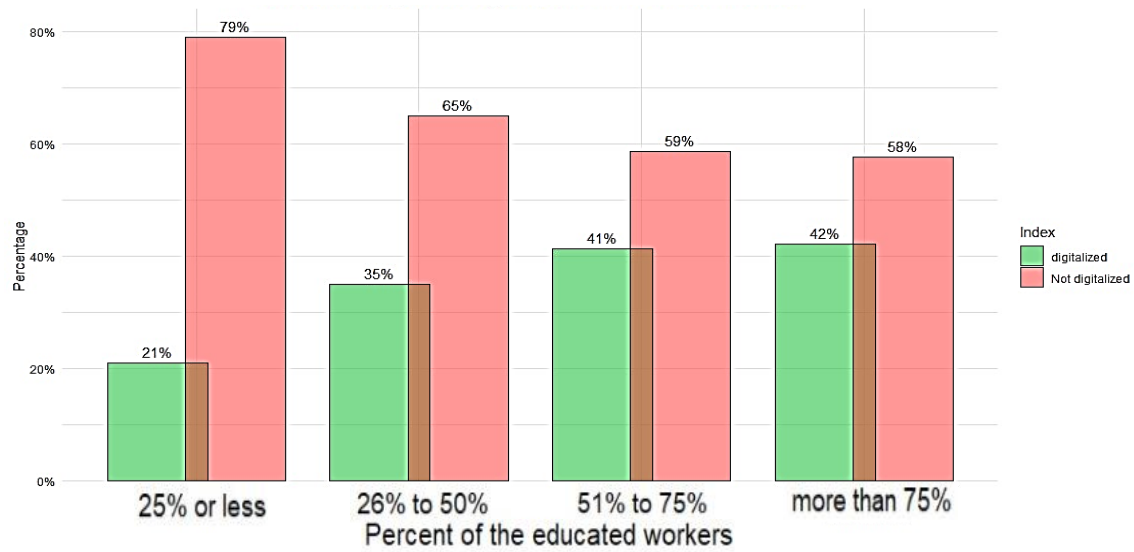


Figure (3.11): The distribution of the DTI by percentage of educated workers.

Figure (3.11) shows that 42% of enterprises with more than 75% of educated workers among their human capital are digitalized. The gap between the percentage of being "digitalized" and "not digitalized" is reduced due to the increase in the percentage of educated workers.

- **The distribution of the Digital Transformation Index by main activity:**

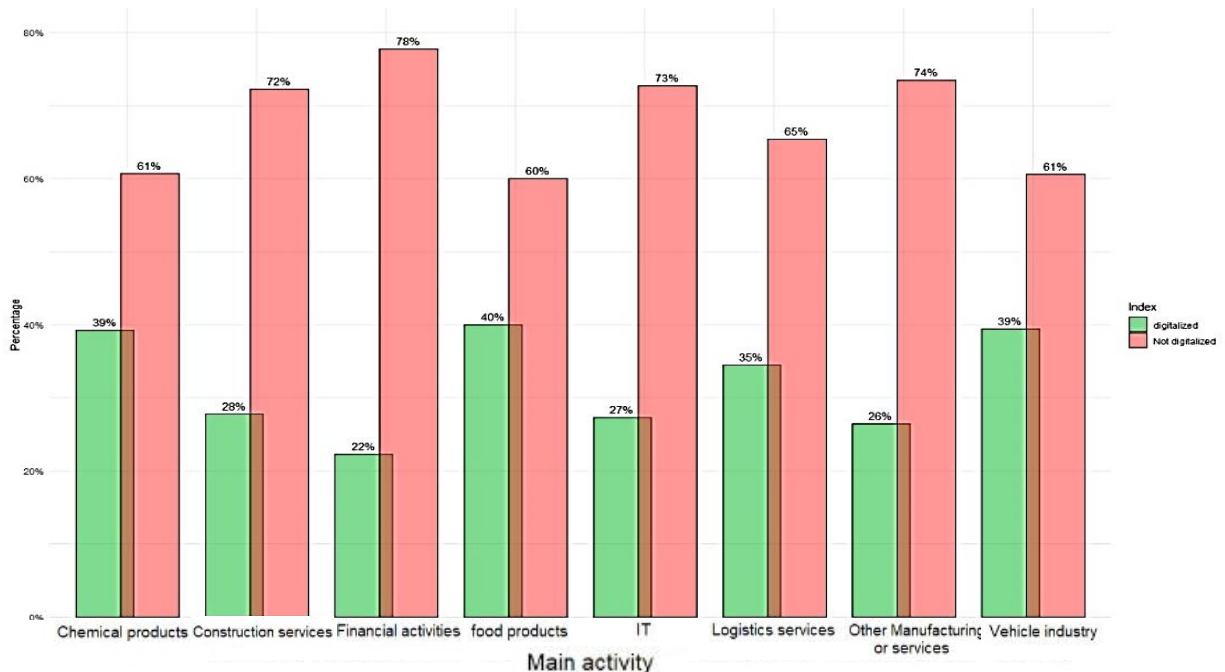


Figure (3.12): The distribution of the DTI by main activity.

Figure (3.12) shows that there is no substantial difference among the main activities of enterprises included in our study.

Chapter (4)

Data Analysis

In order to achieve the objective of this study, we apply the following:

4.1 Digital Transformation Index (DTI):

We conduct the factor analysis on the variables in table (A.1), which are related to the digital transformation.

To ensure the validation of the factor analysis applied, we compute Cronbach alpha³. As shown in Table (4.1), the factor analysis is not valid, and the number of items is not sufficient to apply the factor analysis, as the value of alpha equals 0.019, which is close to 0.

Table (4.1): Cronbach alpha for 16 variables.

Number of items	16
Number of sampled units	948
alpha	0.019

Therefore, we investigate the association among the variables, and this can be shown in table (A.2). From this table, we can see that variables ("using the internet", "digital skills for workers", "cost of the internet", "period of access to the internet", "using the computer" and "number of no internet days") are the only ones that are not associated with other variables. Therefore, we conduct the factor analysis on the variables in table (A.1) except "using internet", "digital skills for workers", "cost of the internet", "period of access to the internet", "using the computer" and "number of no internet days".

See the results in table (4.2), table (4.3) and figure (4.1).

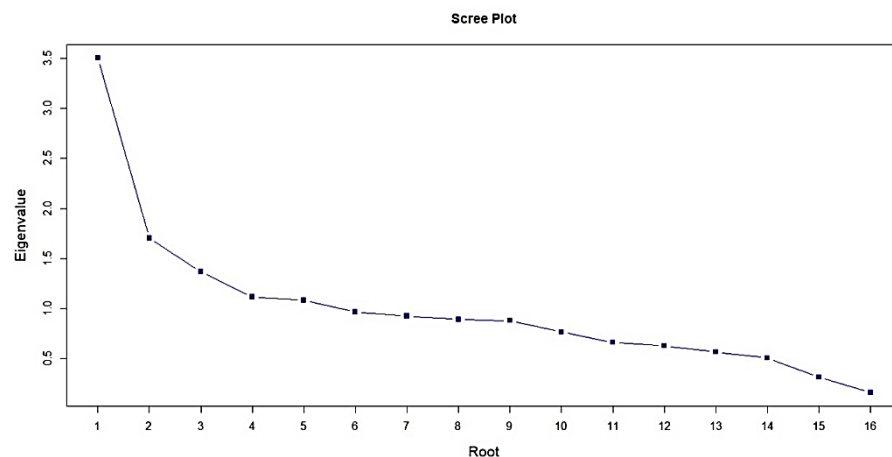


Figure (4.1): Scree plot for 16 variables.

³ **Cronbach alpha** is a function of the variances and covariances of the items to check the sufficiency of these items. If its value is close to 1, it is an indicator of the sufficiency of the items used.

Figure (4.1) shows that the line declines steeply at the beginning and then levels off at the fourth root, which means that we should retain the first three factors.

Table (4.2): Cronbach alpha for 10 variables.

Number of items	10
Number of sampled units	948
alpha	0.75

From table (4.2), we can see that the value of Cronbach alpha, which equals 0.75, is better compared to the previous one in table (4.1), which indicates the validation of the factor analysis and that the items are sufficient to apply the analysis since the Cronbach alpha is close to 1.

Determining the number of retained factors can be done with some approaches. In the first approach, drawing the scree plot on which the y-axis represents the value of the eigenvalue, and the x-axis represents the order of these eigenvalues, the optimal number of retained factors according to this approach is the number of factors before the stage of levelling off after the steep decline of the line, as we can observe in figure (4.2). The second approach is the explained variation. In table (4.3), we can retain only one factor, which explains almost 50% of total variation, as it is acceptable in social science fields. The last approach is the Kaiser-Guttman rule, which retains only the factors whose corresponding eigenvalues are greater than 1 but this approach is not the best approach, as it is an arbitrary approach and results in retaining more factors, as we can see in table (4.3).

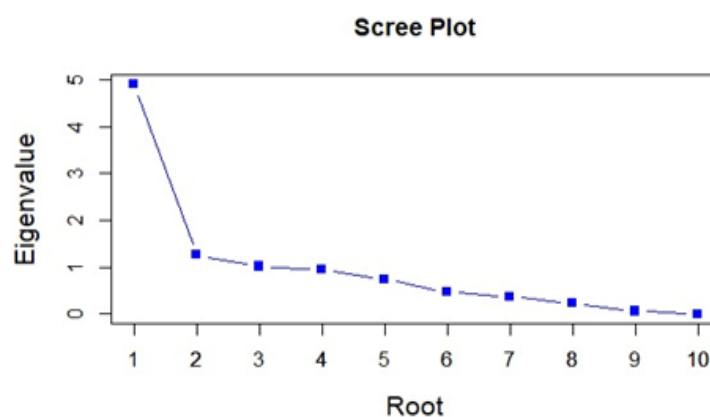


Figure (4.2): Scree plot for 10 Variables.

From figure (4.2), we can see that the line declines steeply in the beginning, then at the second root, it begins to decline gradually, and after that, it has begun to level off, so we retain the factors before this stage, which is the first factor.

Table (4.3): The extracted factors from data.

Factors	Eigenvalues (λ_i)	Proportion of variance	Cumulative proportion of variance
Factor 1	<u>4.91</u>	0.49	0.49
Factor 2	<u>1.26</u>	0.13	0.62
Factor 3	<u>1.01</u>	0.10	0.72
Factor 4	0.95	0.09	0.81
Factor 5	0.75	0.07	0.89
Factor 6	0.47	0.05	0.93
Factor 7	0.37	0.04	0.97
Factor 8	0.23	0.02	0.99
Factor 9	0.06	0.01	1.00
Factor 10	0.00	0.00	1.00

From table (4.3), we can see that the first factor explains almost 50% of the total variation in the data compared to the other factors, which is acceptable for us. Based on the criterion of total explained variation, we retain only the first factor, and this is clear in the table by the **bold format**, but based on the Kaiser-Guttman rule, we should retain the first three factors whose eigenvalues are greater than 1, and this is clear in the table by the underlined format.

We do not depend on the Kaiser-Guttman rule because of the drawbacks that we have mentioned before.

The goal of our study is to construct the digital transformation index. Based on the above criteria and the value of the multiple R square of scores with factors that equals 0.96, we retain only one factor that is compatible with our goal. Then we normalize the scores of the retained factor, which are obtained using the Bartlett factor scores method, so that our indicator ranges from 0 to 1. We find that the closer the DTI value is to 1, the more digitally transformed the enterprise is.

4.2 Cluster analysis results:

In this section, we aim to classify the enterprises into main clusters according to their performance in digital transformation.

We try to cluster using all 16 variables in table (A.1) and then using the 10 categorical variables in table (A.1) to compare the performance of clustering between them to finally use the combination with the best results.

Due to the various types of 16 variables, which are numeric and categorical variables, we normalize the numeric ones to be between 0 and 1, so that the numeric values don't outweigh the categorical values since they will be either 0 or 1, as well as to make good use of the mixed Euclidean distance, which is a combination of the Hamming distance and normal Euclidean distance applied to numeric attributes.

First, to determine the number of clusters, we use the elbow method, as shown below:

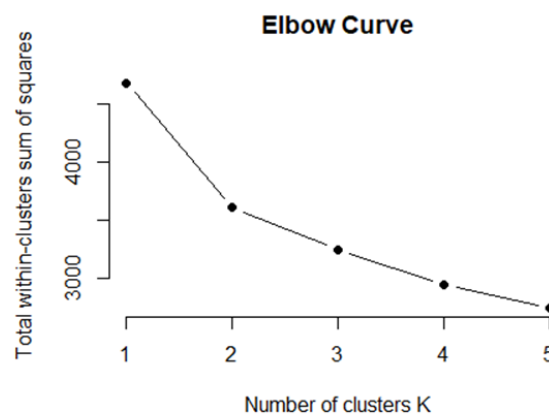


Figure (4.3): Elbow curve for clusters

Although we could have more than five clusters, we only tried to set $k = 5$ for better visibility. We can see that the elbow point happens when the number of clusters is two, where adding more clusters reduces the Within-Cluster Sum of Squares (WCSS) but also increases the sophistication of the model. It is also worth noting that two clusters align with what we want in the next section about binary logistic regression, such that having a binary case is more common in interpretation than having more than two classes.

For 16 variables, we use the k-means algorithm, and for 10 categorical variables, we use the k-modes algorithm to partition the data points into clusters using the hamming distance to finally have the following table:

Table (4.4): Evaluation metrics for clusters.

Metric	16 variables	10 variables
Within cluster sum of squares (WCSS)	3607.55	23.42736
Between clusters sum of squares (BCSS)	4974.05	20810.57
Average Silhouette Score	0.2365883	0.278708

From table (4.4), we can see that the two clusters using the 10 variables have lower WCSS and higher BCSS, which implies the better clustering capability of the 10 variables, where the firms are close to each other within a cluster but the clusters are far from each other. The average silhouette score is relatively high in 10 variables, which indicates that a firm is better clustered within its group and poorly clustered with the others.

After we divide the enterprises into two main clusters, say cluster 1 and cluster 2, we wish to label these clusters based on the summary statistics (i.e., mode for the nominal variables) of the variables used within these clusters, as shown in table (4.5).

Table (4.5): Mode of variables used in clustering.

Variable	Cluster 1	Cluster 2
Having website	Yes	No
Listing in apps	No	No
Online purchase	Yes	No
Online sales	Yes	No
Having online sales	Yes	No
Online inputs	Yes	No
Using smartphones	Yes	No
Digital skills for workers	Yes	Yes
IT person	Yes	No
Social media	Yes	Yes

From table (4.5), we can label cluster 1 as "Digitalized", and label cluster 2 as "Not digitalized".

Validation using the discriminant correspondence analysis:

After applying the cluster analysis to classify the enterprises in Egypt, Morocco and Jordan into "Digitalized" and "Not digitalized" enterprises in table (4.5). We apply the evaluation metrics for clusters in table (4.4) to validate the cluster analysis, and we also apply the discriminant correspondence analysis using the classification table for the same purpose in table (4.6).

Table (4.6): Classification table using the discriminant correspondence analysis.

Predicted (classified)	True		Total
	Digitalized	Not digitalized	
Digitalized	34.75%	0.35%	35.05%
Not digitalized	3.5%	61.4%	64.95%
Total	38.25%	61.75%	100%

Sensitivity	90.8%
Specificity	99.4%
Overall correct classification	96.15%

Table (4.6) shows the results of the classification table of the discriminant correspondence analysis after dividing the dataset into 664 (70%) enterprises in the training dataset and 285 (30%) enterprises in the test dataset. We test the predictive power of the classification on the test data, and the previous table shows that the sensitivity (the percentage of correctly classified enterprises as digitalized enterprises) equals 90.8%, the specificity (the percentage of correctly classified enterprises as not digitalized enterprises) equals 99.4%, and the overall correct classification equals 96.15%, which supports the predictive power of the classification for our clusters.

4.3 Binary logistic model results:

We investigate the factors that affect the Digital Transformation Index in Egypt, Jordan and Morocco. In the previous section, we cluster enterprises as "Digitalized" and "Not Digitalized" due to their performance in digital transformation. Therefore, we fit the binary logistic model for DTI as a dependent categorical variable with a set of variables that are related to the digital transformation and not used in constructing the DTI. We show these variables in table (A.3).

First: Testing the assumptions of the model

Since all variables are categorical, we do not test the assumptions of the binary logistic model that need continuous variables. We investigate the absence of multicollinearity using the Generalized Variance Inflation Factor (GVIF), as follows:

Table (4.7): GVIF for variables in the model.

Variable Name	GVIF
Country	1.681669
Region of the head office	1.046637
Main activity	1.717027
Education of the owner	1.803488
Human capital	1.289272
Using the internet	1.131316
Research and development	1.128765
Cost of the internet	1.204551
Percentage of educated workers	1.587939

From table (4.7), we have noticed that GVIF for all explanatory variables are less than 10. Therefore, the assumption of "Absence of multicollinearity" is satisfied.

Second: Studying the goodness of fit for the model, as follows:

1. Pearson's chi-squared test:

Table (4.8): Pearson's chi-squared test for the model.

Chi-squared statistic	771.68
Degrees of freedom	730
P-value	0.1383

From table (4.8), since p-value is greater than $\alpha = 0.05$, then we do not reject the null hypothesis that assumes that the model is true.

2. ROC Curve:

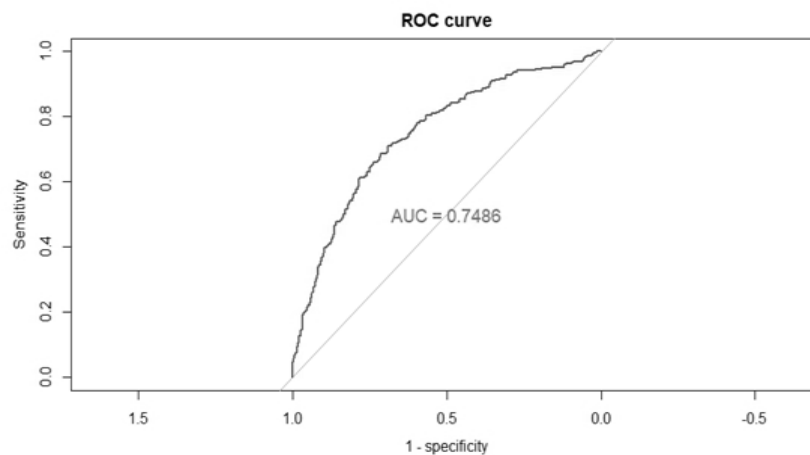


Figure (4.4): ROC curve for the model.

From figure (4.4), the area under the ROC curve is greater than 0.5, which means that our DTI model is better than the intercept model. The best cutoff point equals to 0.3425 based on the Yuden's index.

3. The classification table:

Table (4.9): Classification table for the model.

	True		Total
Predicted (classified)	Digitalized	Not digitalized	
Digitalized	23.8%	19.1%	42.9%
Not digitalized	10.9%	46.2%	57.1%
Total	34.7%	65.3%	100%

Sensitivity	68.6%
Specificity	70.75%
Overall correct classification	70%

From table (4.9), we can conclude that the sensitivity equals 68.6%, which is the percentage of the enterprises that are correctly classified as "Digitalized" and the specificity equals almost 70.75%, which is the percentage of the enterprises that are correctly classified as "Not digitalized". The percentage of overall correct classification equals 70%, then our model has acceptable predictive power.

Third: Interpreting the results of the binary logistic model, as follows:

After applying the Stepwise selection, we obtain these results:

Table (4.10): Results of the binary logistic model.

Variable Name	Coefficients	Odds Ratio	Std.Error	P-Value
Intercept	0.02181	1.0220490	0.81403	0.978626
Region of the head office: (Reference: Rural)				
Urban	0.68452	1.9828245	0.42529	0.107497
Main activity: (Reference: Vehicle Industry)				
Chemical products	-0.02856	0.9718472	0.33241	0.931540
Construction services	-0.91819	0.3992419	0.27592	0.000875***
Financial activities	-1.53835	0.2147355	0.44741	0.000585***
food products	-0.14838	0.8621063	0.24383	0.542832
IT	-1.24460	0.2880575	0.35259	0.000416***
Logistics services	-0.85229	0.4264352	0.24934	0.000630***
Others	-0.93327	0.3932661	0.27604	0.000723***
Education of the owner: (Reference: University or above)				
Illiterate	-0.68244	0.5053842	0.30428	0.024913*
Less than primary	-1.37791	0.2521049	1.08779	0.205261
Primary/Preparatory	-1.42798	0.2397921	0.66194	0.030984*
Secondary Technical/general or Azhar	-0.95309	0.3855477	0.32482	0.003344**
Intermediate Institute/ Higher Institute	-0.40921	0.6641760	0.32118	0.202638
Human capital (Reference: 200 or more)				
Less than 6	-1.54910	0.2124394	0.63131	0.014136*
6 to 49	-1.40671	0.2449482	0.62654	0.024756*
50 to 199	-0.47172	0.6239286	0.69502	0.497322
Introducing new products: (Reference: No)				
Yes	0.24036	1.2717108	0.16633	0.148425
Research and development: (Reference: No)				
Yes	0.91209	2.4895105	0.17335	1.43e-07***
Cost of the internet: (Reference: High)				
Low	-0.50950	0.6007973	0.20584	0.013316*
Medium-H	-0.17988	0.8353678	0.22905	0.432257
Medium-L	-0.01077	0.9892911	0.21213	0.959521
Percentage of educated workers: (Reference: 25% or less)				
26% to 50%	0.47265	1.6042410	0.23311	0.042600*
51% to 75%	0.57481	1.7767869	0.26821	0.032104*
more than 75%	0.82323	2.2778474	0.22327	0.000227***

Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

With 95% confidence, we find that:

- The countries under study have no significant effect on the level of digital transformation.
- There is no significant effect of the region in which the head office is located on the level of digital transformation.
- There is a significant effect of the main activity of the enterprises on the level of digital transformation; as the odds of being digitalized is 60.08%, 78.53%, 71.2%, 57.36% and 60.67% respectively less for activities related to construction, finance, IT, logistics and other activities compared to the vehicle industry, holding other variables constant.
- There is a significant effect of the educational level of the owner on the level of digital transformation; as the odds of being digitalized is 49.46%, 76.02% and 61.45% respectively less for the owner who is illiterate, with a primary or preparatory level and a secondary level compared to those whose educational level is university or above, holding other variables constant. In other words, this means that when the owner of an enterprise is highly educated, the enterprise tends to be digitalized.
- There is a significant effect of the enterprise's human capital on the level of digital transformation; as the odds of being digitalized is 78.76% and 75.51% respectively less for enterprises whose number of paid workers is less than 6 and between 6 and 49 compared to enterprises whose number of paid workers is more than 200, holding other variables constant, which means that when an enterprise has more paid workers among its human capital, the enterprise tends to be digitalized.
- Introducing new products or services into the main market has no significant effect on the level of digital transformation.
- There is a significant effect of spending on research and development on the level of digital transformation; as the odds of being digitalized is 148.95% more for enterprises that spend on research and development compared to those who don't, holding other variables are constant, which reflects the importance of spending on research and development.

- There is a significant effect of the cost of the internet as a share of total cost on the level of digital transformation; as the odds of being digitalized is 39.92% less for enterprises that have low internet cost compared to those who have high internet cost, holding other variables constant.
- There is a significant effect of the percentage of educated workers on the level of digital transformation; as the odds of being digitalized is 60.42%, 77.68% and 127.78% more for the share of highly educated workers, which ranges between 26%:50%, 51%:75% and more than 75% compared to the share which is 25% or less, holding other variables constant, which shows the importance of educated workers on the level of digital transformation.
- The number of competitors in the main market has no significant effect on the level of digital transformation.
- The percentage of workers whose work requires computer usage has no significant effect on the level of digital transformation; as using only computers is not related to digital transformation but digitization, as we have shown the difference between two terms before in the chapter 1.
- The number of days with no internet access or power outage has no significant effect on the level of digital transformation.
- The age of the enterprise and the period of access to the internet have no significant effect on the level of digital transformation.

4.4 Decision tree results:

Before we apply the most common machine learning technique for classification and predicting whether the enterprise is "Digitalized" or "Not digitalized", we divide the data into training and test datasets. The training dataset contains 70% of the data, and the test dataset contains 30% of the data. We first apply the decision tree model to the training dataset, as shown in figure (4.5)⁴.

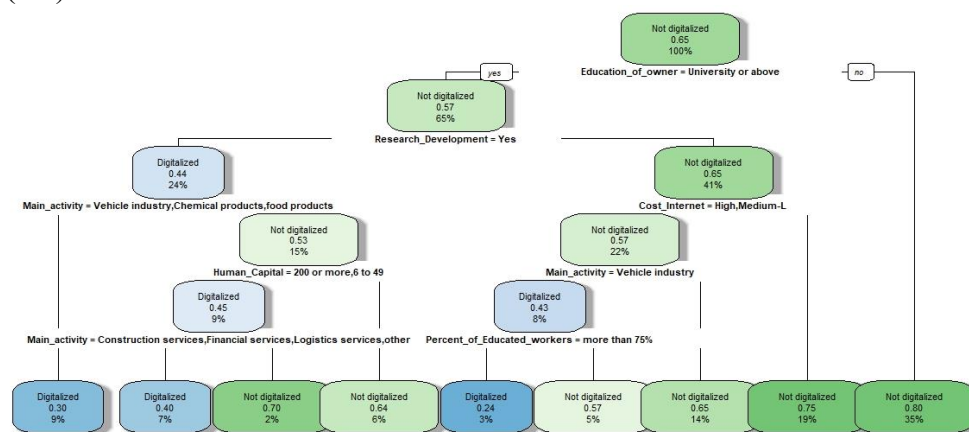


Figure (4.5): Decision tree for the training data.

From figure (4.5), we conclude that:

At the root node, we have 100% of enterprises in the data before splitting them based on the entropy with a probability of being "Not digitalized" equal to 0.65 and a probability of being "Digitalized" equal to 0.35.

Then this node is split into two nodes based on the education of the owner. One node contains 35% of the enterprises whose owners have less education than those at the university stage; their enterprises tend to be "Not digitalized" with a probability equal to 0.8, and the other node contains 65% of the enterprises whose owners are highly educated at the university or above stage; their enterprises tend to be "Not digitalized" with a probability equal to 0.57.

Then the previous node is split based on research and development into two nodes. One node contains 41% of the enterprises that do not spend on research and development; they tend to be "Not digitalized" with a probability equal to 0.65, and the other node contains 24% of the enterprises that spend on research and development; they tend to be "Digitalized" with a probability equal to 0.44.

⁴ The color blue or light blue represents the enterprises that tend to be "Digitalized" and the color green or light green represents the enterprises that tend to be "Not digitalized".

Then 41% of the enterprises that do not spend on research and development are split into several decision nodes and leaves based on more than one variable. One node contains 19% of the enterprises whose cost of the internet is low or medium-high; they tend to be "Not digitalized" with a probability equal to 0.75. Another node contains 22% of the enterprises whose cost of the internet is medium-low or high; they tend to be "Not digitalized" with a probability equal to 0.57. Then the enterprises whose cost of the internet is medium or high are split based on the main activity into 14% of the enterprises that do not have any activity in the vehicle industry; they tend to be "Not digitalized" with a probability equal to 0.65, and 8% of the enterprises that have an activity in the vehicle industry are "Digitalized" with a probability equal to 0.43.

Then enterprises in the vehicle industry are split into two nodes by the percentage of educated workers. One node contains 5% of the enterprises that have less than 75% of educated workers; they tend to be "Not digitalized" with a probability equal to 0.57, and the other node contains 3% of the enterprises that have more than 75% of educated workers; they tend to be "Digitalized" with a probability equal to 0.24.

24% of the enterprises that spend on research and development are split into several decision nodes and leaves based on more than one variable, one node contains 9% of the enterprises whose main activity in the vehicle industry, chemical, or food products; they tend to be "Digitalized" with a probability equal to 0.3, another node contains 15% of the enterprises that have another main activity; they tend to be "Not digitalized" with a probability equal to 0.53, and then these enterprises are split into two nodes based on the human capital; such that 6% of the enterprises that have (less than 6) or (from 6 to 199) workers; they tend to be "Not digitalized" with a probability equal to 0.64, and 9% of the enterprises that have (from 6 to 49) workers or (200 or more) workers; they tend to be "Digitalized" with a probability equal to 0.45.

The enterprises that have (200 or more) workers are split into two nodes based on the main activity, one node contains 7% of the enterprises whose main activity in construction, financial, logistics or any other services; they tend to be "Digitalized" with a probability equal to 0.4, and the other node contains 2% of the enterprises whose main activity in the vehicle industry, chemical or food products; they tend to be "Not digitalized" with a probability equal to 0.7.

Cross validation:

Table (4.11): Importance of variables in splitting data.

Variable	Overall
Percentage of educated workers	27.187527%
Main activity	22.885368%
Research and development	20.895093%
Education of the owner	16.926351%
Cost of the internet	9.244600%
Human capital	9.080489%

From table (4.11), we conclude that:

- The most important variable in deciding whether an enterprise is digitalized or not is the percentage of educated workers in the enterprise.
- The two least important variables in determining whether an enterprise is digitalized or not are both the cost of the internet and human capital.

Table (4.12): Classification table for the decision tree.

Prediction	Reference (True)		Total
	Digitalized	Not digitalized	
Digitalized	12.3%	22.8%	35.1%
Not digitalized	6.6%	58.3%	64.9%
Total	18.9%	81.1%	100%

Sensitivity	65.08%
Specificity	71.88%
Overall correct classification	70.6%

From table (4.12), the percentage of overall correct classification of the decision tree model by applying it to the test data equals 70.6%, which means that the decision tree model has acceptable predictive power in classifying enterprises as digitalized or not.

Chapter (5)

Conclusion and Recommendations

A. Conclusions:

The aim of the study is to construct a multidimensional quantitative index to measure the level of digital transformation of Egyptian, Jordanian, and Moroccan enterprises in 2022. First, a Digital Transformation Index is created using the factor analysis technique. Then enterprises are clustered based on their performance in digital transformation. After that, we investigate the main factors affecting the level of digital transformation and test whether or not there is a significant difference in the level of digital transformation among Egypt, Jordan, and Morocco.

From the different analyses conducted, we come up with:

- The Digital Transformation Index gives a certain value between 0 and 1 for each enterprise according to its performance in digital transformation.
- Enterprises in our study can be clustered into two groups: "digitalized" or "not digitalized" based on their performance in digital transformation.
- The main factors affecting the level of digital transformation are the human capital of enterprises, the educational level of the owner and the percentage of educated workers.
- For the educational level of the owner, there is a significant effect on the level of digital transformation, which means that when the owner of the enterprise is highly educated, the enterprise tends to be digitalized.
- For the percentage of educated workers, there is also a significant effect on the level of digital transformation, which means that when the enterprise has a large percentage of educated workers, it tends to be digitalized.
- For the human capital of an enterprise, the enterprise is more likely to be digitalized when it has more paid workers.
- The main activity of the enterprise, its age, spending on research and development, introducing new product to the main market, the number of its competitors in the main market, and its interest in digital skills for workers are associated with the Digital Transformation Index but they all have no significant effect on the level of digital transformation.

B. Recommendations:

These results support the need to pay attention to education because of its effective role in the digital transformation process and its subsequent impact on the desired economic growth rate in those countries. As well as interest in enhancing the values of digital transformation among individuals and institutions through initiatives.

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Appendix

Table (A.1): Variables used in constructing the DTI.

Name in data	Definition	Short name	Type
Cor11	Does your firm have its own website in (last month) 2022?	Having website	Categorical
Cor12	Is the firm listed on any app or website – e.g. a food delivery app?	Listing in apps	Categorical
Cor13	Did your firm make any online purchases in (last month) 2022?	Online purchase	Categorical
Cor14	Did your firm make any online sales in (last month) 2022?	Online sales	Categorical
Cor16	Did your firm use the Internet in (last month) 2022?	Using the internet	Categorical
Fir8_2	How long has your establishment had access to high-speed internet? (Year)	Period of access to the internet	Quantitative
Fir10	What was the cost of the internet as a share of your establishment's total cost last month?	Cost of the internet	Quantitative
Fir12	Did your establishment have online sales (last month) in 2022?	Having Online sales	Categorical
Fir14	Did your establishment purchase inputs (any materials that be used for the workflow) online (last month) in 2022)?	Online inputs	Categorical
Fir16	Have your firm employees used the smartphones to conduct online transactions for business?	Using smartphones	Categorical
Fir17	What is the percentage of workers in the firm their work requires computer usage?	Using the computer	Quantitative
Fir18	Do digital skills play a role in your decision to appoint managerial staff?	Digital skills for managerial staff	Categorical
Fir19	Do digital skills play a role in your decision to recruit workers?	Digital skills for workers	Categorical
Fir20	Does the Firm have IT person (whether full time, part time)?	IT Person	Categorical
Fir22	Are you using the social media for business purposes?	Social media	Categorical
Fir25	What is the number of days with no online services in last month?	Days of no online services	Quantitative

Table (A.2): Association matrix

	Cor11	Cor12	Cor13	Cor14	Cor16	Fir12	Fir14	Fir16	Fir22	Fir20	Fir18	Fir19	Fir25	Fir10	Fir17	Fir8_2
Cor11	1	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.13 *	0.88 *	0.09 *	0.00	0.00
Cor12		1	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.03	0.04	0.04	0.73 *	0.45 *	0.01
Cor13			1	0.00	0.02	0.00	0.00	0.00	0.00	0.00	0.00	0.11 *	0.90 *	0.30 *	0.00	0.02
Cor14				1	0.17 *	0.00	0.00	0.00	0.00	0.00	0.00	0.08 *	0.43 *	0.25 *	0.42 *	0.52 *
Cor16					1	0.10 *	0.03	0.00	0.09 *	0.04	0.09 *	0.03	0.35 *	0.08 *	0.55 *	0.05 *
Fir12						1	0.00	0.00	0.00	0.00	0.02	0.05 *	0.87 *	0.33 *	0.31 *	0.46 *
Fir14							1	0.00	0.00	0.00	0.01	0.06 *	0.24 *	0.10 *	0.42 *	0.12 *
Fir16								1	0.00	0.00	0.00	0.15 *	0.98 *	0.47 *	0.04	0.23 *
Fir22									1	0.02	0.00	0.10	0.25 *	0.03	0.02	0.13 *
Fir20										1	0.00	0.15 *	0.09 *	0.56 *	0.00	0.00
Fir18											1	0.31 *	0.15 *	0.28 *	0.00	0.00
Fir19												1	0.02	0.45 *	0.12 *	0.25 *
Fir25													1	0.03	0.02	0.37 *
Fir10														1	0.24 *	0.78 *
Fir17															1	0.00
Fir8_2																1

We measure the association among categorical variables using the Pearson's chi-squared test for independence, and we measure the association among quantitative variables using the Pearson's Correlation.

Since the normality assumption of quantitative variables is rejected at $\alpha = 0.05$, we use non-parametric measures to assess the correlations among quantitative and qualitative variables, such that:

Mann-Whitney test for correlation among quantitative and qualitative variables (except for IT Person because it has more than two classes).

Kruskal-Wallis test for correlation among quantitative variables and IT Person.

Values inside the table represent the p-value of previous tests.

(*) represents that two variables are independent at ($\alpha = 0.05$).

Table (A.3): Variables used in estimating the binary logistic model.

Name in data	Definition	Short name	Type
country	Country (<i>Egypt/Jordan/Morocco</i>)	Country	Categorical
cor9	Is the head office of this establishment located at Urban/Rural? (<i>Rural/Urban</i>)	Region of the head office	Categorical
cor7	Establishment's main activity (<i>Vehicle industry/Chemical products/Construction services/ Financial activities/food products/IT/Logistics services/ Other Manufacturing or services</i>)	Main activity	Categorical
fir6_b	What is the last educational level that the owner completed successfully? (<i>Illiterate/Less than primary/Primary or preparatory/ Secondary/ Intermediate institute or Higher institute/University or above</i>)	Education of owner	Categorical
cor5	Currently, How many workers were working in your firm including yourself (paid)? (<i>Less than 6 / 6 to 49 / 50 to 199 / 200 or more</i>)	Human capital	Categorical
fir40	What is the number of competitors for your main product/service in the main market? (<i>1 to 5 / 6 to 20 / 21 to 100 / 101 or more</i>)	Number of competitors	Categorical
fir44	Has your establishment introduced new products/services, or introduced any innovation? (<i>Yes / No</i>)	Introducing new products	Categorical
fir10	What was the cost of the internet as a share of your establishment's total cost last month? (<i>Low / Medium-Low / Medium-High / High</i>)	Cost of the internet	Categorical
fir7	On average, what is the share of Highly educated workers does this establishment? (<i>25% or less/ 26% to 50% / 51% to 75% / more than 75%</i>)	Percentage of educated workers	Categorical
fir17	What is the percentage of workers in the firm their work requires computer usage? (<i>Low / Medium-Low / Medium-High / High</i>)	Using the computer	Categorical
fir25	What is the number of days with no internet access (interruption) in (last month)? (<i>Low / Medium / High</i>)	Number of no internet days	Categorical
cor10	In what year did your establishment begin operations? (<i>Low / Medium-Low / Medium-High / High</i>)	Age of company	Categorical
fir45	Has your establishment spent on research and development during last 3 years? (<i>Yes / No</i>)	Research and development	Categorical
fir8_2	How long has your establishment had access to high speed internet?(year) (<i>Short / Medium / Long</i>)	Period of access to the internet	Categorical
fir19	Do digital skills play a role in your decision to recruit workers? (<i>Yes / No</i>)	Digital skills for workers	Categorical
fir24	On Average, What is the number of power outage in (last month) 2022? (<i>Low / Medium / High</i>)	Number of power outage days	Categorical